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High lithium electroactivity of hierarchical porous rutile TiO_2 nanorod microspheres

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ABSTRACT

Here we report on the hierarchical porous rutile TiO_2 nanorod micospheres as an anode material for lithium-ion batteries. The resulting hierarchical porous rutile TiO_2 nanorod microspheres possessed much higher reversible capacity, cycling stability and rate capability than nanosized rutile TiO_2 previously reported in the literatures. These good electrochemical performances may be attributed to the facile diffusion of Li^+ ions from outside through the porous channels into the TiO_2 nanorods in the microspheres and the high electrode–electrolyte contact area offered by hierarchical porous microspheres with a large specific surface area.

1. Introduction

Lithium-ion batteries have emerged as power sources for modern electronics because of their high-energy storage density, long cycle life, and no memory effect. Present commercial lithium-ion batteries use $LiCoO_2$ as positive-electrode material, and graphite as negative-electrode material [1]. The graphite electrodes have some disadvantages, such as an initial capacity loss, structural deformation and electrical disconnection [2]. In order to avoid these drawbacks, transition metal oxides such as WO_3, MoO_3, and TiO_2, have attracted more and more attention [3–5].

The titanium oxides with one-dimensional (1D) nanostructures bring several important features including large specific surface area and pore volume as well as the dimensions in the two directions perpendicular to long axis of the particles comparable to the mean-free-path length of electrons [6]. As the potential electrode materials, materials with large specific surface area and pore volume are often desirable. Meanwhile, the one-dimensional (1D) structure will facilitate the diffusion of lithium-ions into TiO_2. Therefore, one-dimensional TiO_2 is a promising electrode material for lithium-ion battery.

Anatase TiO_2 is generally considered to be the most electroactive Li insertion host [6–8]. While Li insertion into rutile is usually reported to be less favorable at room temperature [9–18]. It is known that bulk rutile can only accommodate a negligible amount of Li (<0.1 Li per TiO_2 unit) at room temperature [16–19]. Interestingly, it was recently reported that nanosized rutile TiO_2 could reversibly accommodate 0.5 Li ($Li_{0.5}TiO_2$), comparable to that of anatase [20–22]. This new finding initiates new attempts to synthesize nanostructured rutile TiO_2 with lithium electroactivity.

In this communication, we report that hierarchical porous rutile TiO_2 nanorod microspheres can be inserted up to 0.73 Li ($Li_{0.73}TiO_2$, 246 mAh/g) during the first discharge at a C/10 charge/discharge rate. More importantly, these hierarchical porous microspheres possess an excellent cycle performance. Their stabilized capacity is as high as 192 mAh/g after 30 cycles. Furthermore, the hierarchical porous rutile TiO_2 nanorod microspheres also exhibit high capacity at higher charge/discharge rate, suggesting their excellent rate capability.

2. Experimental section

2.1. Sample preparation

The hierarchical porous rutile TiO_2 nanorod microspheres were synthesized according to our previous study [23]. The molar ratio of $TiCl_4$, ethanol and water was 2:20:280.

2.2. Characterization

X-ray powder diffraction patterns were obtained using a Philips MPD-18801 diffractometer. Scanning electron microscopy measurements were performed used a JSM-5600 SEM. Transmission electron microscopy study was carried out on a Philips CM-120. The nitrogen adsorption and desorption isotherms were measured using Micrometrics ASAP2010 system.

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2.3. Electrochemical measurements

The TiO_2 electrodes were prepared by mixing 80% TiO_2 nanorod microspheres, 10% acetylene black and 10% poly(tetrafluoroethylene) by weight with 2-propanol to form a paste. The paste was roll-pressed into ca. 0.1 mm thick film and finally fixed onto a nickel net. The testing cells had a typical three-electrode construction using lithium foils as both counter electrodes and reference electrodes, and 1 M $LiPF_6$ dissolved in ethylene carbonate (EC), dimethyl carbonate (DMC) and ethylene methyl carbonate (EMC) (1:1:1, v/v/v) as the electrolyte. The cells were assembled in an argon-filled glove box.

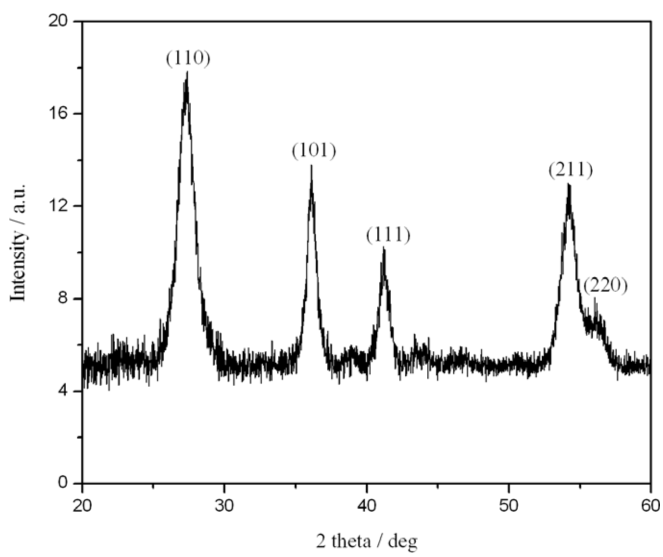

Fig. 1. XRD patterns of the as-prepared sample.

3. Results and discussion

XRD diffraction of the as-prepared sample reveal a pure tetragonal rutile phase (JCPDS No. 21-1276, space group: $P4_2/mnm$ (1 3 6)) (Fig. 1). No peaks of anatase or brookite phase are detected, indicating the high purity of the products. SEM images show that a lot of well-defined spheres from 200 to 600 nm in size are observed in the as-prepared sample (Fig. 2a). Fig. 2b displays that these spheres possess rough surface and are radically covered with acicular particles, looking like urchins. The image of a broken sphere in Fig. 2c confirms that these urchin-like particles are composed of nanorods with acicular tips. These nanorods are 3–5 nm in diameter and 40–60 nm in length, respectively. The TEM image reveals the nanorods on the edges of spheres (Fig. 2d). The nitrogen adsorption–desorption isotherms of TiO_2 nanorod microspheres are of two distinct capillary condensation steps, revealing the hierarchical porous system of the sample (Fig. 3). The Brunauer–Emmett–Teller (BET) surface area of the sample is 166.8 m^2/g.

Fig. 4 displays the initial and second charge/discharge curves as well as cycle behavior (inset) of the electrode made of the hierarchical porous rutile nanorod microspheres at a C/10 charge/discharge rate. The initial charge and discharge capacity of the electrode are 356 and 246 mAh g^{-1} (Fig. 4), respectively, indicating a reversible efficiency as high as 69% for Li insertion and extraction in the initial discharge–charge curves. The initial discharge capacity was significantly higher than that of the previously reported nanosized rutile TiO_2 (about 200 mAh/g) [20–22] and TiO_2 nanotubes (about 230 mAh/g) [8,24]. The enhanced capacity may be ascribed to shorter transport lengths for both electronic and Li^+ transport in the hierarchical porous nanorod microspheres. In addition, the hierarchical porous microspheres with a large specific surface area could increase the electrode–electrolyte contact area, which facilitates the lithium-ions insertion and extraction. It can be observed from Fig. 4 that there is a voltage plateau at approximately 1.4 V and a short voltage plateau near 1.05 V in the first discharge curve. Similar voltage plateaus were also observed in other

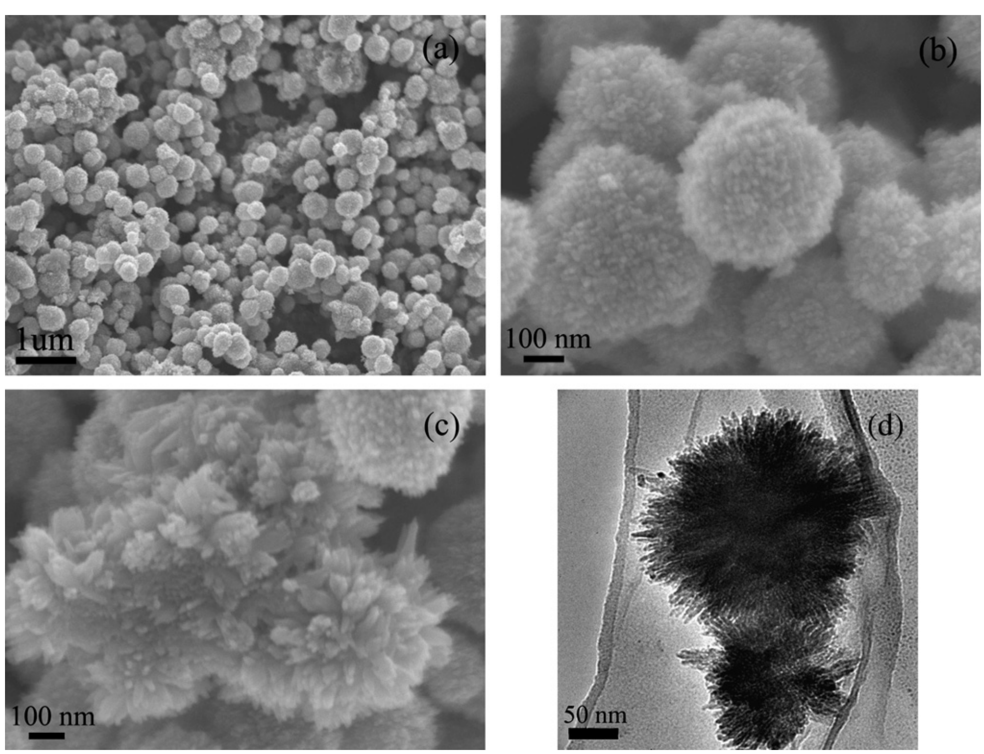

Fig. 2. SEM images (a–c) and TEM image (d) of the as-prepared sample.

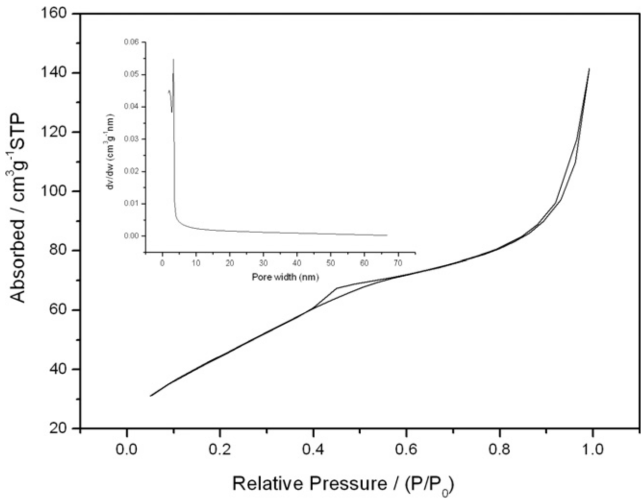

Fig. 3. Nitrogen adsorption–desorption isotherm and pore size distribution curve (inset) of the as-prepared sample.

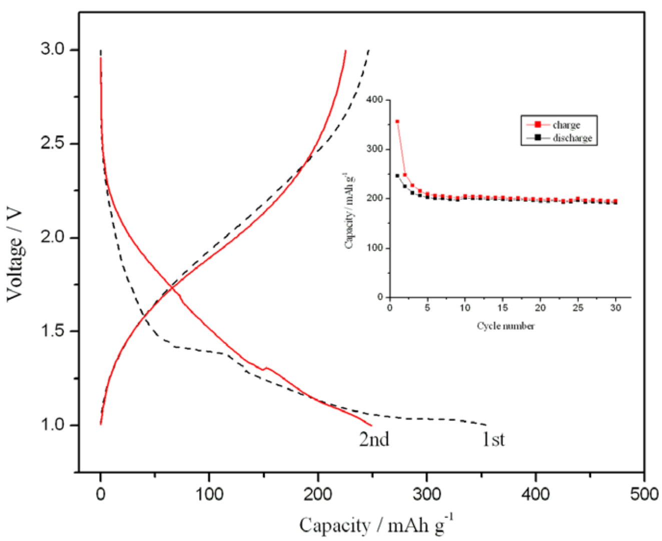

Fig. 4. The Initial and second discharge–charge curves and cycle behavior (inset) of the electrode made of the hierarchical porous rutile nanorod microspheres at a C/10 charge/discharge rate.

literature [20]. This large irreversible capacity of about 100 mAh g^{-1} in the first cycle was previously reported by Maier and coworkers [20]. They demonstrated that the structure was changed irreversibly only at deeper Li insertion (between 1.4 and 1.0 V).

In the subsequent discharge processes, the distinct voltage plateau at 1.4 V was not very clear, and the short voltage plateau near 1.05 V for discharging disappeared. The irreversible formation of a 'nanocomposite' of crystalline grains and amorphous regions might be the reason for the sloped behavior at the subsequent discharge process [25–27]. In the second cycle, the insertion and extraction capacities were observed to be 248 and 225 mAh g^{-1}, respectively. The irreversible capacity of 23 mAh g^{-1} was much smaller than that (about 100 mAh g^{-1}) of the first cycle. This can be attributed to three reasons. The first reason is that the irreversible Li insertion sites (related to the short voltage plateau near 1.05 V for discharging) are filled completely in the first cycle. The second is that the trace water is consumed gradually in the initial several cycles. And the third is that the residual lithium in TiO_2 could improve

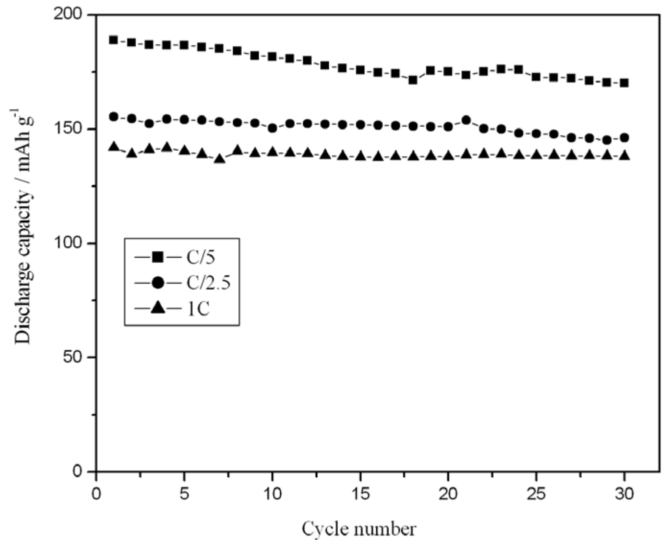

Fig. 5. Discharge capacity with respect to cycle number for the hierarchical porous rutile nanorod microspheres electrodes at different charge–discharge rate.

the electric conductivity [15]. Thus, the irreversible capacity decreased and the coulombic efficiency increased in the subsequent cycles. The inset of Fig. 4 reveals that the hierarchical porous microspheres exhibit an excellent cycle performance and their stabilized capacity is as high as 192 mAh g^{-1} after 30 cycles. This stabilized capacity was much higher than that of TiO_2 nanotubes (about 120 mAh/g) [24].

Another excellent property of the hierarchical porous rutile nanorod microspheres is their high-rate capability. Fig. 5 shows discharge capacity with respect to cycle number for the hierarchical porous rutile TiO_2 nanorod microspheres electrode at different charge–discharge rate. The discharge specific capacities of 189, 155, 142 mAh g^{-1} are obtained at C/5, C/2.5, and 1C rates, respectively. The excellent high-rate performance might also be resulted from the transport advantages of this special nanostructure, such as shorter transport lengths for both electronic and Li^+ transport. In addition, the hierarchical porous microspheres with a large specific surface area could increase the electrode–electrolyte contact area, and thus decrease the current density per unit surface area, leading to higher charge/discharge rate.

4. Conclusions

In summary, we have demonstrated that the hierarchical porous rutile TiO_2 nanorod microspheres possess higher reversible capacity, cycling stability and rate capability than nanosized rutile TiO_2 previously reported in the literatures. These hierarchical porous rutile TiO_2 nanorod microspheres can be inserted up to 0.73 Li ($Li_{0.73}TiO_2$, 246 mAh g^{-1}) during the first discharge at a C/10 charge/discharge rate. After 30 cycles, their stabilized capacity is as high as 192 mAh g^{-1}. Moreover, the hierarchical porous rutile TiO_2 nanorod microspheres also exhibit high capacity at higher charge/discharge rate. The excellent electrochemical performances make the hierarchical porous rutile TiO_2 nanorod microspheres a promising anode material for high-power lithium-ion batteries.

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